1 PINN Burgers

[ ]:
```
import os
os.environ['CUDA_VISIBLE_DEVICES']='0'

import torch
import torch.nn as nn
import numpy as np
import pandas as pd
from scipy.io import loadmat
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker

import time

from sklearn.metrics import mean_squared_error, mean_absolute_error


plt.rcParams['font.sans-serif']=['Times New Roman'] #
plt.rcParams['axes.unicode_minus']=False

np.random.seed(1234)
torch.manual_seed(1234)


device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print('running on: %s'%(device))

#
base_dir = os.getcwd()
print(f"Current working directory: {base_dir}")
# base_dir = os.path.dirname(os.path.abspath(__file__))
output_path = os.path.join(base_dir, 'Hom2', 'Output', 'PINN_pred.mat')


#
file = loadmat('Hom2\\burgers_shock.mat')
```

```
base_x = file['x']
base_t = file['t']
base_u = file['usol']
print('x_size:', base_x.shape, 't_size:', base_t.shape, 'u_size:', base_u.shape)
```

```
running on: cuda
Current working directory: d:\MyDocs\Workspace\SciML_Course_Homework
x_size: (256, 1) t_size: (100, 1) u_size: (256, 100)
```

[2]:
```
#   tensor
def to_tensor(x):
    return torch.tensor(x, dtype=torch.float32)

#
def part_t(x=np.linspace(-1, 1, 256),
           t=np.linspace(0,0.9,100),
           u=None,
           time_slices=[0, 0.2, 0.4, 0.6, 0.8]):
    t=t.flatten()
    id_t = []
    for j in time_slices:
        id = np.argwhere(t==j).flatten()
        id_t += id.tolist()
    u_slices = u[:,id_t]
    print(u_slices.shape)
    return u_slices
```

[3]:
```
#
def predict(model, x, t):
    X, T = np.meshgrid(x, t)

    XT = np.hstack([X.flatten()[:,None], T.flatten()[:,None]])
    XT = to_tensor(XT)

    U_pred = model(XT).detach().numpy()
    return U_pred.reshape(len(t), len(x))
```

1.1

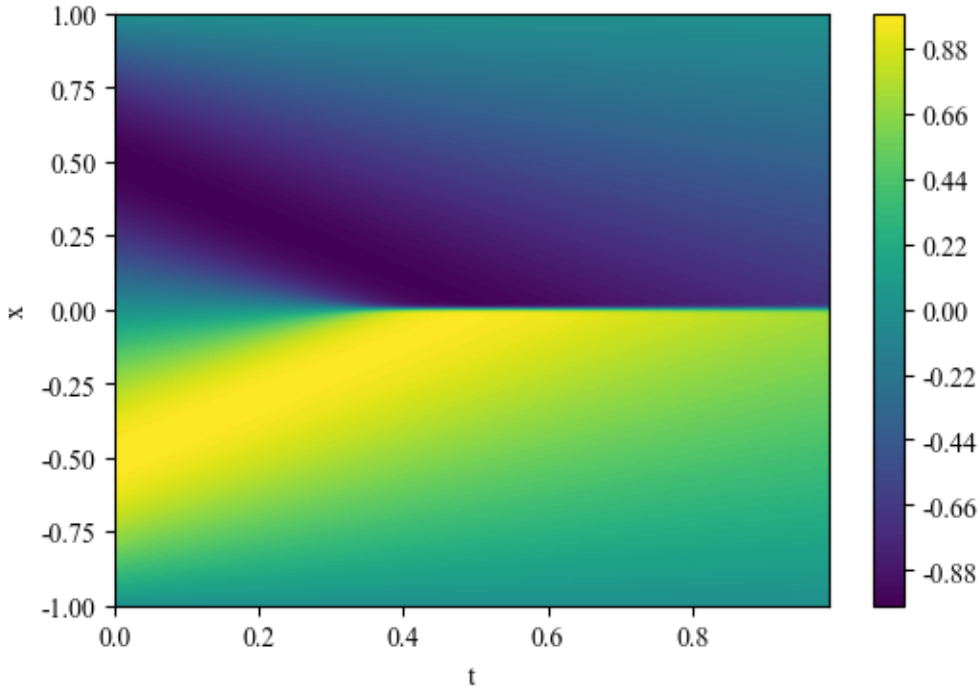
[4]:
```
#
def plot_sol_2d(model=None, x=np.linspace(-1, 1, 256),
                t=np.linspace(0, 0.9, 100), u=None, title=None):
    X, T = np.meshgrid(x, t)
    U = predict(model, x, t) if u is None and model is not None else u

    plt.figure(figsize=(6, 4))
    plt.contourf(T, X, U, 100, cmap='viridis')
```

```
    plt.colorbar()
    plt.xlabel('t')
    plt.ylabel('x')
    plt.title(title)
    plt.show()

#
def plot_sol_3d(model=None, x=np.linspace(-1, 1, 256),
                t=np.linspace(0, 0.9, 100), u=None):
    X, T = np.meshgrid(x, t)
    U = predict(model, x, t) if u is None and model is not None else u

    fig = plt.figure(figsize=(6, 4))
    ax = fig.add_subplot(111, projection='3d')
    ax.plot_surface(X, T, U, cmap='viridis')

    ax.set_xlabel('x')
    ax.set_ylabel('t')
    ax.set_zlabel('u(t, x)')
    ax.set_title('Solution of Burgers Equation on (x, t) Plane')
    plt.show()

#  t (x,u)
def plot_sol_curve(model=None,
                   x=np.linspace(-1, 1, 256),
                   t=np.linspace(-1, 0.9, 100),
                   u=None,
                   time_slices=[0, 0.2, 0.4, 0.6, 0.8],
                   label=None):  #

    if len(time_slices) >= 2:
        plt.figure(figsize=(6, 4))
        #       t=0
        u_initial = -np.sin(np.pi * x)
        plt.plot(x, u_initial, 'k:', label='Exact t=0', linewidth=2, alpha=0.7)


    if u is None and model is not None:

        for i, t_val in enumerate(time_slices):
            #
            t_vals = np.full_like(x, t_val)[:, None]
            XT_slice = np.hstack((x[:, None], t_vals))
            XT_tensor = to_tensor(XT_slice)
            #
            u_pred = model(XT_tensor).detach().numpy()
```

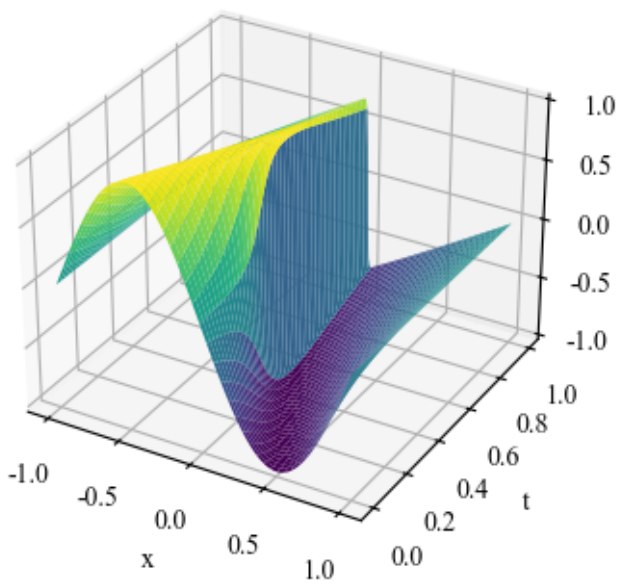
```
            #
            plt.plot(x, u_pred,
                    label=f't={t_val}' if label is None else label,
                    linewidth=2)

    else:
        u_pred = part_t(x=x, t=t, u=u,time_slices=time_slices)
        for i, t_val in enumerate(time_slices):
            part_u = u_pred[:,i]
            plt.plot(x, part_u,
                    label=f't={t_val}' if label is None else label,
                    linewidth=2)


    #
    plt.xlabel("x", fontsize=14)
    plt.ylabel("u(t,x)", fontsize=14)
    plt.title("Burgers Equation at "+", ".join(map(str,time_slices)))
    plt.legend(fontsize=12, loc='best')
    plt.grid(True, alpha=0.15)
    plt.xlim([-1, 1])
    # plt.tight_layout()
    if len(time_slices) >=2:
        plt.show()
```

1.2

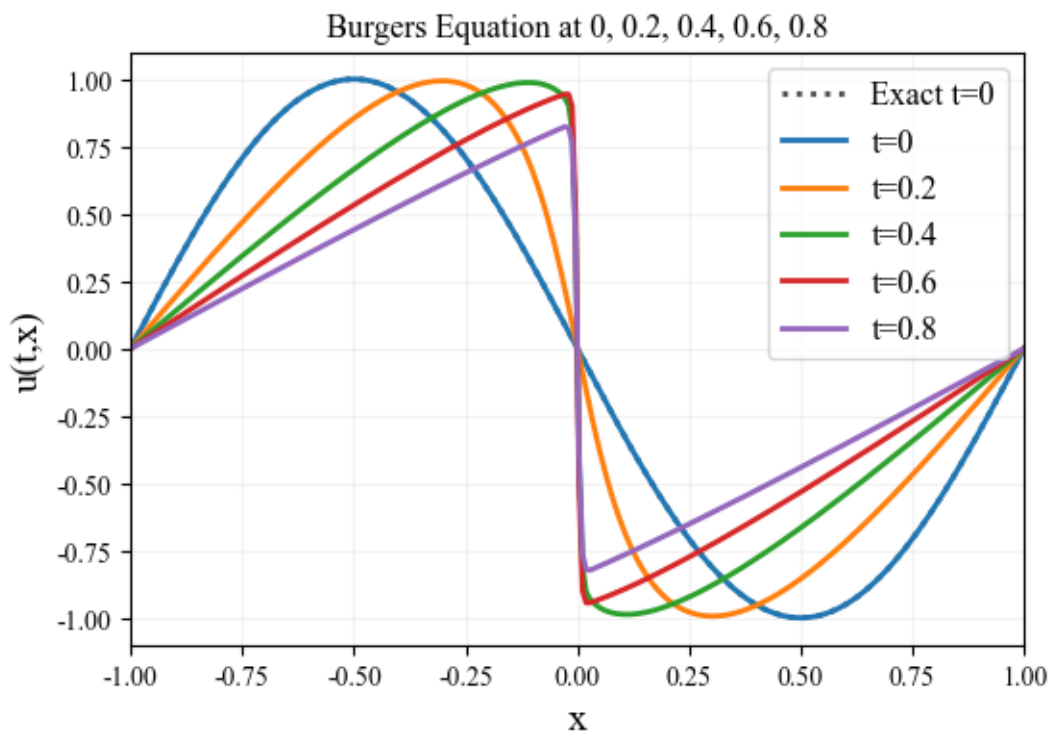
[5]:
```
plot_sol_2d(x=base_x, t=base_t, u=base_u.T)
plot_sol_3d(x=base_x, t=base_t, u=base_u.T)
plot_sol_curve(x=base_x, t=base_t, u=base_u)
```



Solution of Burgers Equation on (x, t) Plane




```
(256, 5)
```



1.3 PINN

[6]:
```
class PINN(nn.Module):
    def __init__(self, layers=[2,30,30,1]):
        super(PINN, self).__init__()
        L = len(layers)
        self.layers = layers
        self.activation = nn.Tanh()

        self.linears = nn.ModuleList()
        for i in range(1, L):
            layer = nn.Linear(layers[i-1], layers[i])
            self.linears.append(layer)

            nn.init.xavier_normal_(layer.weight)
            nn.init.zeros_(layer.bias)


    def forward(self, x):
        A = x
        for i in range(len(self.linears) - 1):
            A = self.activation(self.linears[i](A))
```

```
        Y = self.linears[-1](A)
        return Y


    def pdenn(self, x, t, nu=0.01/np.pi):
        x = x.clone().detach().requires_grad_(True)
        t = t.clone().detach().requires_grad_(True)

        u = self.forward(torch.cat([x,t], dim=1))

        u_t = torch.autograd.grad(u, t, torch.ones_like(u), retain_graph=True,␣
↪create_graph=True)[0]
        u_x = torch.autograd.grad(u, x, torch.ones_like(u), retain_graph=True,␣
↪create_graph=True)[0]
        u_xx = torch.autograd.grad(u_x, x, torch.ones_like(u),␣
↪retain_graph=True, create_graph=True)[0]

        f = u_t + u*u_x - nu*u_xx
        return f
```

1.4

[7]:
```
#
def ic(x):
    return -np.sin(np.pi * x)
#
def bc(t):
    return np.zeros_like(t)
```

1.5

[8]:
```
#
nu = 0.01 / np.pi

#
def training_data(N_f=8000, N_bc=200, N_ic=200):
    # PDE
    x_f = np.random.uniform(-1, 1, (N_f, 1))
    t_f = np.random.uniform(0, 0.9, (N_f, 1))

    #    t=0
    x_ic = np.random.uniform(-1, 1, (N_ic, 1))
    t_ic = np.zeros_like(x_ic)
    u_ic = ic(x_ic)

    #    x=-1,1
```

```
    t_bc = np.random.uniform(0, 0.9, (N_bc, 1))

    x_bc_l = -np.ones((N_bc, 1))
    x_bc_r = np.ones((N_bc, 1))
    u_bc = bc(t_bc)

    x_bc = np.vstack([x_bc_l, x_bc_r])
    t_bc = np.vstack([t_bc, t_bc])
    u_bc = np.vstack([u_bc, u_bc])
    # u_bc = np.zeros_like(x_bc)

    return x_f, t_f, x_ic, t_ic, u_ic, x_bc, t_bc, u_bc

#
def add_data(model, N_new=500):
    x_f = np.random.uniform(-1, 1,(N_new,1))
    t_f = np.random.uniform(0,0.9, (N_new,1))

    x_f = to_tensor(x_f).to(device)
    t_f = to_tensor(t_f).to(device)

    f = model.pdenn(x_f, t_f, nu=0.01/np.pi)
    err = np.abs(f.detach().cpu().numpy())

    idx = np.argsort(err[:,0])[-100:]
    x_add = x_f[idx]
    t_add = t_f[idx]
    return x_add, t_add
```

1.6

[9]:
```
def train_model(rar=False, layers=[2,30,30,1], nmax=30000):
    #
    if rar :
        print('    RAR')
    datas = training_data() if rar==False else training_data(N_f=2000)
    datas_tensor = map(to_tensor, datas)
    datas_device = map(lambda x: x.to(device), datas_tensor)
    x_f, t_f, x_ic, t_ic, u_ic, x_bc, t_bc, u_bc = datas_device

    #
    model =PINN(layers).to(device)

    optimizer = torch.optim.Adam(model.parameters(), lr=1e-3)
    mse = nn.MSELoss()

    #
```

```
ls_loss_f = []
ls_loss_ic = []
ls_loss_bc = []
loss_results = []
added_points={'x':[],'t':[]}

#
start_time = time.perf_counter()
for n in range(nmax+1):
    # RAR
    if rar and n%500 ==0:
        x_add, t_add = add_data(model)

        added_points['x'] += x_add.detach().cpu().numpy().flatten().tolist()
        added_points['t'] += t_add.detach().cpu().numpy().flatten().tolist()

        x_add = to_tensor(x_add)
        t_add = to_tensor(t_add)
        x_f = torch.cat([x_f, x_add], dim=0)
        t_f = torch.cat([t_f, t_add], dim=0)
        print('# of training data: %d'%(x_f.shape[0]))

    optimizer.zero_grad()

    # PDE
    f_pred = model.pdenn(x_f, t_f, nu=0.01/np.pi)
    loss_f = mse(f_pred, torch.zeros_like(f_pred))

    #
    u_ic_pred = model(torch.cat([x_ic, t_ic], dim=1))
    loss_ic = mse(u_ic_pred, u_ic)

    #
    u_bc_pred = model(torch.cat([x_bc, t_bc], dim=1))
    loss_bc = mse(u_bc_pred, u_bc)

    loss = loss_f + loss_ic + loss_bc

    loss.backward()
    optimizer.step()

    if n % 500 == 0:
        print(f"epoch {n}, Loss: {loss.item():.3e}")
        ls_loss_f.append(loss_f.item())
        ls_loss_ic.append(loss_ic.item())
        ls_loss_bc.append(loss_bc.item())
        loss_results.append(loss.item())
```

```
    stop_time = time.perf_counter()
    print('Duration time is %.3f seconds'%(stop_time - start_time))

    return model.cpu(), ls_loss_f, ls_loss_ic, ls_loss_bc, loss_results,␣
 ↪added_points

default_nmax = train_model.__defaults__[2]
default_n = np.array([i for i in range(0,default_nmax+1,500)])
```

[10]:
```
def rar_model(layers=[2,30,30,1]):
    #
    datas = training_data(N_f=2000)
    datas_tensor = map(to_tensor, datas)
    datas_device = map(lambda x: x.to(device), datas_tensor)
    x_f, t_f, x_ic, t_ic, u_ic, x_bc, t_bc, u_bc = datas_device

    def build_dataset(x, x_add):
        return torch.cat([x, x_add], dim=0)

    #
    model = PINN(layers).to(device)

    optimizer = torch.optim.Adam(model.parameters(), lr=1e-3)
    mse = nn.MSELoss()

    n=0
    nmax = 5
    err = 1.

    n_r_max = 2000

    start_time = time.perf_counter()
    while n <= nmax and err > 1.0e-4:
        print()
        print('Loop: %d'%(n+1))
        if n == 0:
            x_f = x_f
            x_f = t_f
        else:
            x_f = build_dataset(x_f, x_add)
            t_f = build_dataset(t_f, t_add)

        n_r = 0
        loss = 1.0
```

```
        #
        while n_r <= n_r_max and loss >= 1.03-6:
            optimizer.zero_grad()

            # PDE
            f_pred = model.pdenn(x_f, t_f, nu=0.01/np.pi)
            loss_f = mse(f_pred, torch.zeros_like(f_pred))

            #
            u_ic_pred = model(torch.cat([x_ic, t_ic], dim=1))
            loss_ic = mse(u_ic_pred, u_ic)

            #
            u_bc_pred = model(torch.cat([x_bc, t_bc], dim=1))
            loss_bc = mse(u_bc_pred, u_bc)

            loss = loss_f + loss_ic + loss_bc

            loss.backward()
            optimizer.step()

            n_r += 1
            if n_r%100 == 0:
                print('Steps: %d, Loss: %.3e'%(n_r, loss))
        n += 1

        print('# of training data: %d'%(x_f.shape[0]))

        err = np.mean(np.absolute(f_pred.detach().cpu().numpy()))
        print('Mean error of residual: %.5f'%(err))

        #
        x_add, t_add = add_data(model)
        x_add = to_tensor(x_add)
        t_add = to_tensor(t_add)

        print(f"Loss after training: {loss.item():.3e}")

    stop_time = time.perf_counter()
    print('Duration time is %.3f seconds'%(stop_time - start_time))

    return model.cpu()
```

1.7

[11]:
```
# ===    ===
loss_name = ['loss_f', 'loss_ic', 'loss_bc', 'total_loss']

def plot_loss(loss, title='Training Loss Curve'):
    fig, ax = plt.subplots(figsize=(6,4))
    ax.plot(default_n, loss)

    ax.set_yscale('log')
    ax.yaxis.set_major_locator(ticker.LogLocator(base=10))
    ax.yaxis.set_minor_locator(ticker.NullLocator())

    ax.xaxis.set_major_locator(ticker.MultipleLocator(5000))
    ax.xaxis.set_minor_locator(ticker.MultipleLocator(2500))

    # formatter = ticker.ScalarFormatter(useMathText=True)
    # formatter.set_powerlimits((4,4))
    # ax.xaxis.set_major_formatter(formatter)

    plt.xlabel('Step')
    plt.ylabel('Loss')
    plt.grid(ls='-', alpha=0.15)
    plt.title(title)
    plt.legend()
    plt.show()

def plot_loss_dict(loss_dict, title='Training Loss Curve'):
    fig, ax = plt.subplots(figsize=(6,4))
    for key, loss in loss_dict.items():
        ax.plot(default_n, loss, label=key)

    ax.set_yscale('log')
    ax.yaxis.set_major_locator(ticker.LogLocator(base=10))
    ax.yaxis.set_minor_locator(ticker.NullLocator())

    ax.xaxis.set_major_locator(ticker.MultipleLocator(5000))
    ax.xaxis.set_minor_locator(ticker.MultipleLocator(2500))

    plt.xlabel('Step')
    plt.ylabel('Loss')
    plt.grid(ls='-', alpha=0.15)
    plt.title(title)
    plt.legend()
    plt.show()
```

1.8

[12]:
```
# ===  ===

def evaluate(model):
    x_test = torch.tensor(base_x, dtype=torch.float32)
    t_test = torch.tensor(base_t, dtype=torch.float32)
    U_pred = predict(model, x_test, t_test)

    mse = mean_squared_error(base_u.T, U_pred)
    mae = mean_absolute_error(base_u.T, U_pred)

    return U_pred, mse, mae
```

1.9

1.9.1

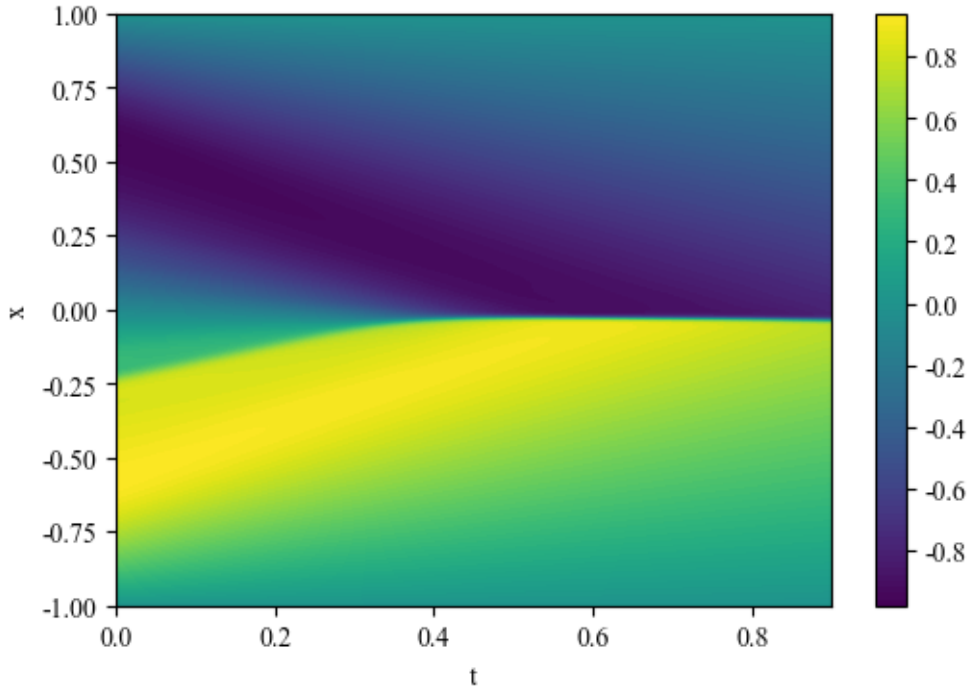
[13]:
```
pro = train_model()
```

```
d:\Software\Python\Python\Lib\site-packages\torch\autograd\graph.py:841:
UserWarning: Attempting to run cuBLAS, but there was no current CUDA context!
Attempting to set the primary context… (Triggered internally at C:\actions-run
ner\_work\pytorch\pytorch\pytorch\aten\src\ATen\cuda\CublasHandlePool.cpp:270.)
  return Variable._execution_engine.run_backward(  # Calls into the C++ engine
to run the backward pass
```

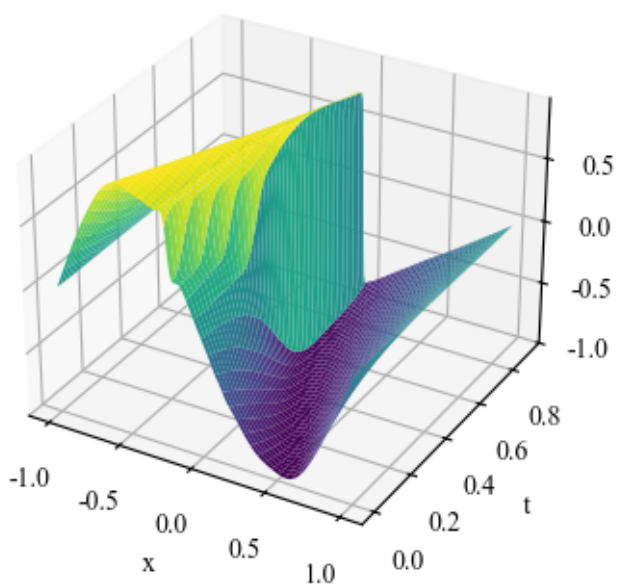
```
epoch 0, Loss: 4.668e-01
epoch 500, Loss: 1.285e-01
epoch 1000, Loss: 1.010e-01
epoch 1500, Loss: 8.988e-02
epoch 2000, Loss: 8.486e-02
epoch 2500, Loss: 8.105e-02
epoch 3000, Loss: 7.682e-02
epoch 3500, Loss: 7.223e-02
epoch 4000, Loss: 6.787e-02
epoch 4500, Loss: 6.449e-02
epoch 5000, Loss: 6.231e-02
epoch 5500, Loss: 6.034e-02
epoch 6000, Loss: 5.805e-02
epoch 6500, Loss: 5.575e-02
epoch 7000, Loss: 5.174e-02
epoch 7500, Loss: 4.932e-02
epoch 8000, Loss: 4.788e-02
epoch 8500, Loss: 4.692e-02
epoch 9000, Loss: 4.618e-02
epoch 9500, Loss: 4.553e-02
epoch 10000, Loss: 4.491e-02
epoch 10500, Loss: 4.423e-02
```

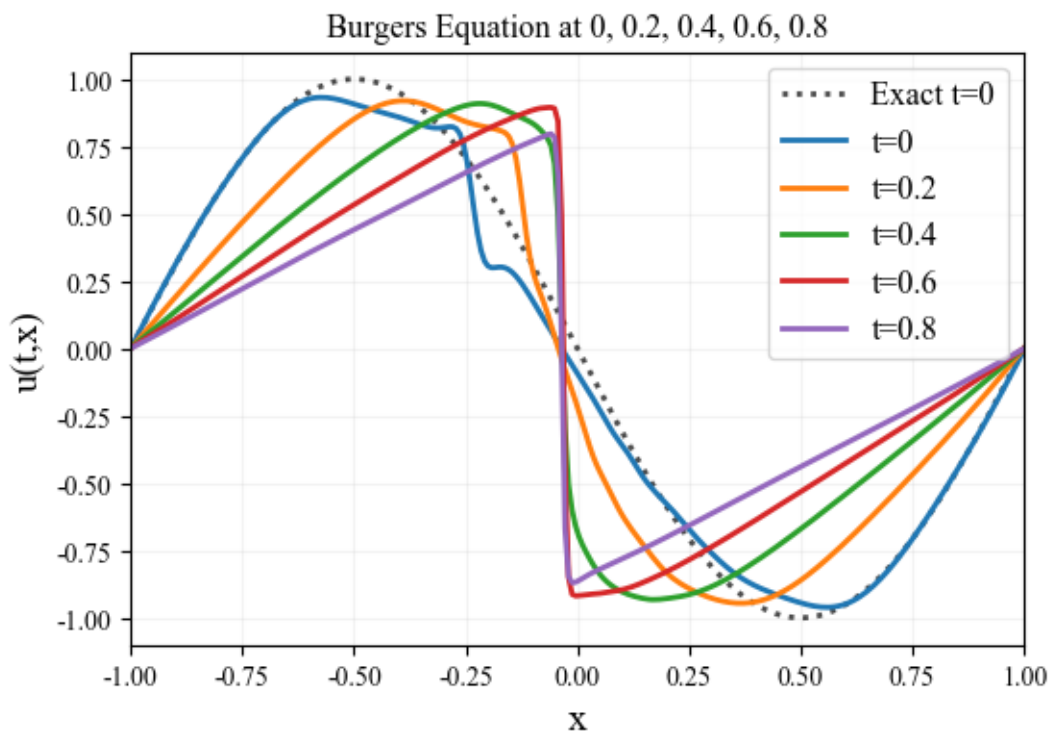
```
epoch 11000, Loss: 4.340e-02
epoch 11500, Loss: 4.238e-02
epoch 12000, Loss: 4.122e-02
epoch 12500, Loss: 3.992e-02
epoch 13000, Loss: 3.863e-02
epoch 13500, Loss: 3.870e-02
epoch 14000, Loss: 3.595e-02
epoch 14500, Loss: 3.448e-02
epoch 15000, Loss: 3.301e-02
epoch 15500, Loss: 3.166e-02
epoch 16000, Loss: 3.043e-02
epoch 16500, Loss: 2.992e-02
epoch 17000, Loss: 2.837e-02
epoch 17500, Loss: 2.728e-02
epoch 18000, Loss: 2.608e-02
epoch 18500, Loss: 2.510e-02
epoch 19000, Loss: 2.430e-02
epoch 19500, Loss: 2.280e-02
epoch 20000, Loss: 2.167e-02
epoch 20500, Loss: 2.054e-02
epoch 21000, Loss: 1.950e-02
epoch 21500, Loss: 1.609e-02
epoch 22000, Loss: 1.272e-02
epoch 22500, Loss: 1.204e-02
epoch 23000, Loss: 1.152e-02
epoch 23500, Loss: 1.108e-02
epoch 24000, Loss: 1.160e-02
epoch 24500, Loss: 1.041e-02
epoch 25000, Loss: 1.006e-02
epoch 25500, Loss: 9.588e-03
epoch 26000, Loss: 9.231e-03
epoch 26500, Loss: 8.889e-03
epoch 27000, Loss: 8.619e-03
epoch 27500, Loss: 8.507e-03
epoch 28000, Loss: 8.048e-03
epoch 28500, Loss: 7.828e-03
epoch 29000, Loss: 7.631e-03
epoch 29500, Loss: 7.456e-03
epoch 30000, Loss: 7.300e-03
Duration time is 350.576 seconds
```

[14]:
```
model = pro[0]

##
plot_sol_2d(model)
plot_sol_3d(model)
plot_sol_curve(model)
```

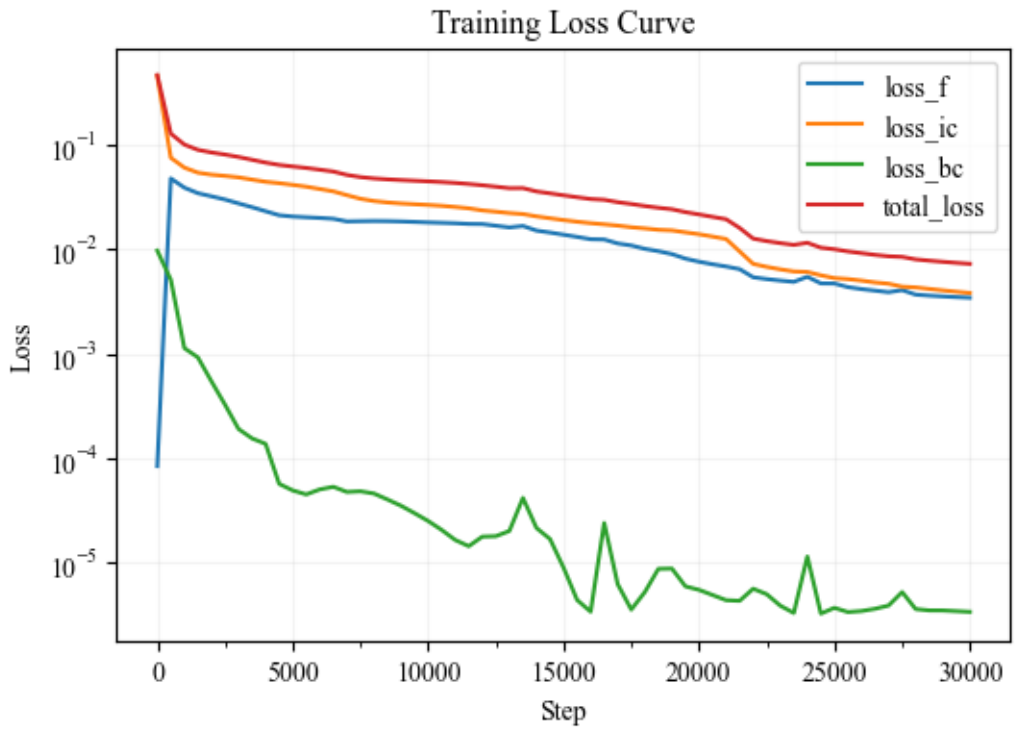


Solution of Burgers Equation on (x, t) Plane



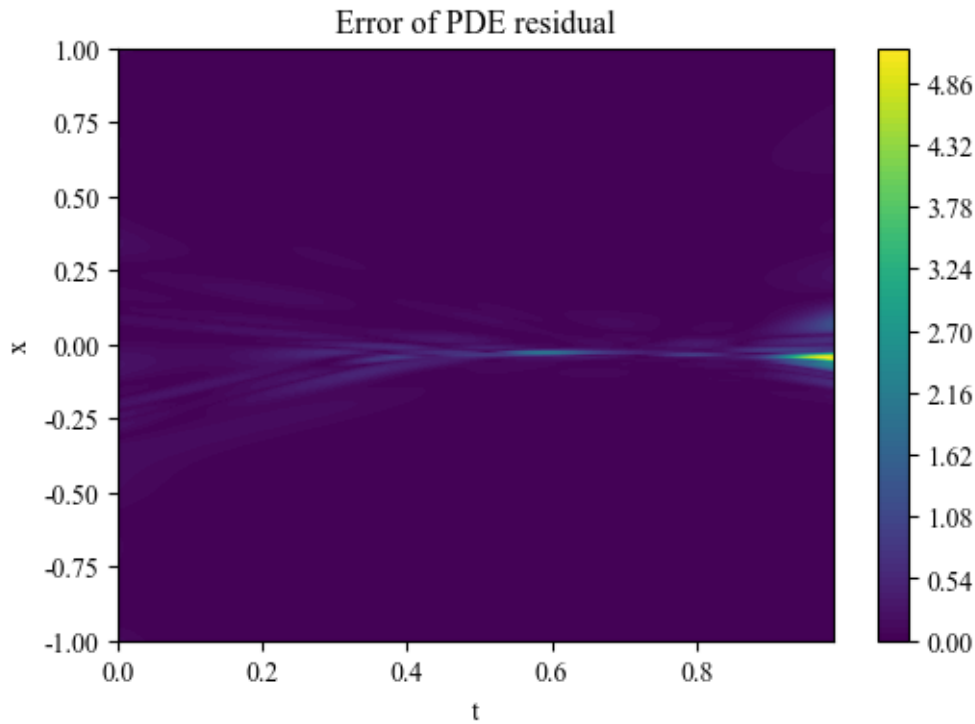


[15]:
```
#

loss = pro[1:-1]
loss_dict = dict(zip(loss_name,loss))

plot_loss_dict(loss_dict)
```



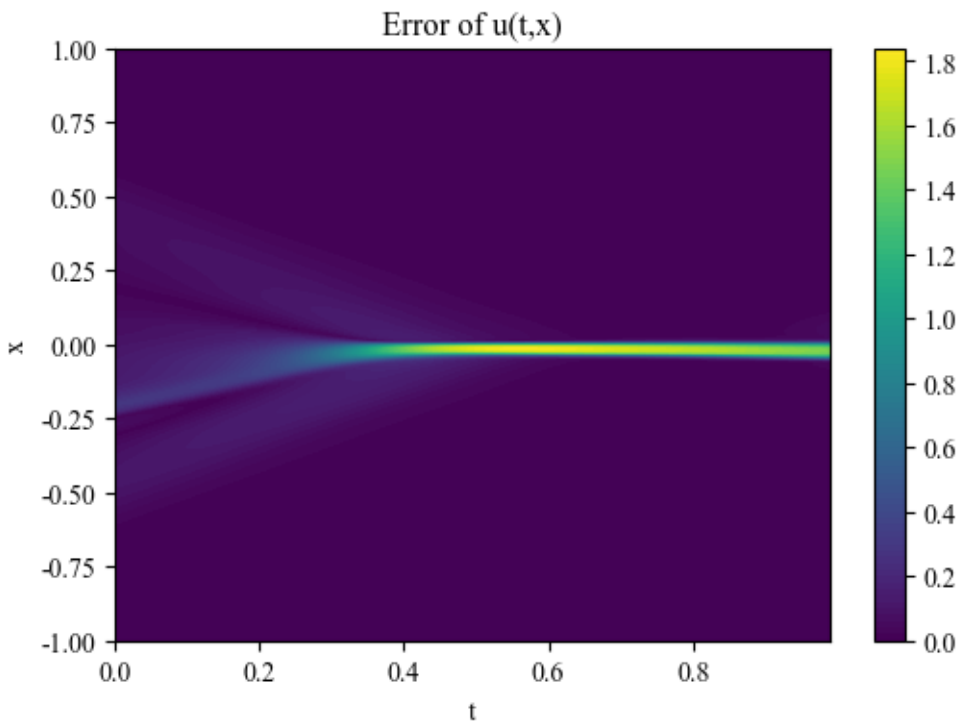
[16]:
```
#

x_test, t_test = base_x, base_t
X_, T_ = np.meshgrid(x_test,t_test)
X_, T_ = X_.reshape(-1,1), T_.reshape(-1,1)

f_pred = model.pdenn(to_tensor(X_),to_tensor(T_)).detach()

mse_pde = mean_squared_error(f_pred, torch.zeros_like(f_pred))
mae_pde = mean_absolute_error(f_pred, torch.zeros_like(f_pred))
print(f'mse_pde:{mse_pde:.5e}, mae_pde:{mae_pde:.5e}')

f_pred = np.abs(f_pred.reshape(len(t_test),len(x_test)).numpy())

plot_sol_2d(x=x_test, t=t_test, u=f_pred,title='Error of PDE residual')
```

```
mse_pde:1.98259e-02, mae_pde:4.09038e-02
```



[17]:
```
#

u_pred, mse, mae = evaluate(model)
print(u_pred.shape)
print(f"MSE:{mse:.5e}, MAE:{mae:.5e}")

err = np.abs(u_pred-base_u.T)
plot_sol_2d(x=base_x, t=base_t, u=err, title='Error of u(t,x)')
```

```
(100, 256)
MSE:2.86210e-02, MAE:4.04942e-02
```



1.9.2

[18]:
```
pro_rar = train_model(rar=True)
```

```
    RAR
# of training data: 2100
epoch 0, Loss: 5.921e-01
```

```
C:\Users\xin\AppData\Local\Temp\ipykernel_28088\2558606513.py:3: UserWarning: To
copy construct from a tensor, it is recommended to use
sourceTensor.detach().clone() or
sourceTensor.detach().clone().requires_grad_(True), rather than
torch.tensor(sourceTensor).
  return torch.tensor(x, dtype=torch.float32)
```

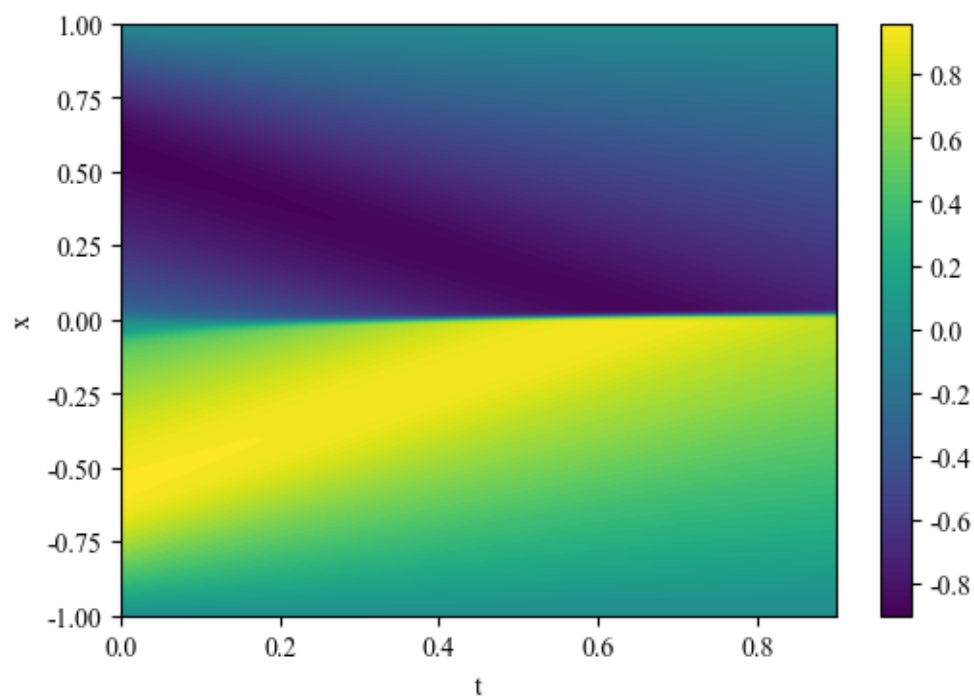
```
# of training data: 2200
epoch 500, Loss: 1.407e-01
# of training data: 2300
epoch 1000, Loss: 1.199e-01
# of training data: 2400
epoch 1500, Loss: 1.051e-01
# of training data: 2500
epoch 2000, Loss: 9.634e-02
# of training data: 2600
```

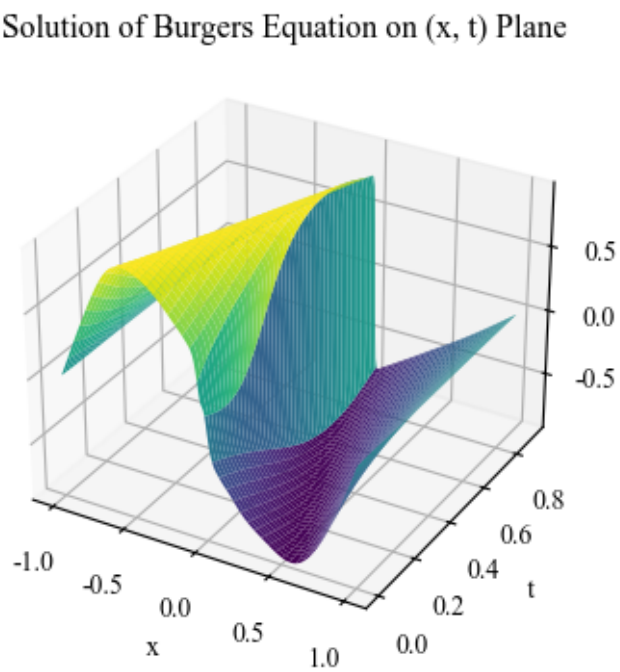
```
epoch 2500, Loss: 9.047e-02
# of training data: 2700
epoch 3000, Loss: 8.462e-02
# of training data: 2800
epoch 3500, Loss: 8.096e-02
# of training data: 2900
epoch 4000, Loss: 7.692e-02
# of training data: 3000
epoch 4500, Loss: 7.253e-02
# of training data: 3100
epoch 5000, Loss: 6.712e-02
# of training data: 3200
epoch 5500, Loss: 5.425e-02
# of training data: 3300
epoch 6000, Loss: 4.862e-02
# of training data: 3400
epoch 6500, Loss: 4.543e-02
# of training data: 3500
epoch 7000, Loss: 4.372e-02
# of training data: 3600
epoch 7500, Loss: 4.245e-02
# of training data: 3700
epoch 8000, Loss: 4.842e-02
# of training data: 3800
epoch 8500, Loss: 3.906e-02
# of training data: 3900
epoch 9000, Loss: 3.764e-02
# of training data: 4000
epoch 9500, Loss: 3.627e-02
# of training data: 4100
epoch 10000, Loss: 3.436e-02
# of training data: 4200
epoch 10500, Loss: 3.225e-02
# of training data: 4300
epoch 11000, Loss: 2.986e-02
# of training data: 4400
epoch 11500, Loss: 2.764e-02
# of training data: 4500
epoch 12000, Loss: 2.563e-02
# of training data: 4600
epoch 12500, Loss: 2.366e-02
# of training data: 4700
epoch 13000, Loss: 2.171e-02
# of training data: 4800
epoch 13500, Loss: 1.986e-02
# of training data: 4900
epoch 14000, Loss: 1.858e-02
# of training data: 5000
```

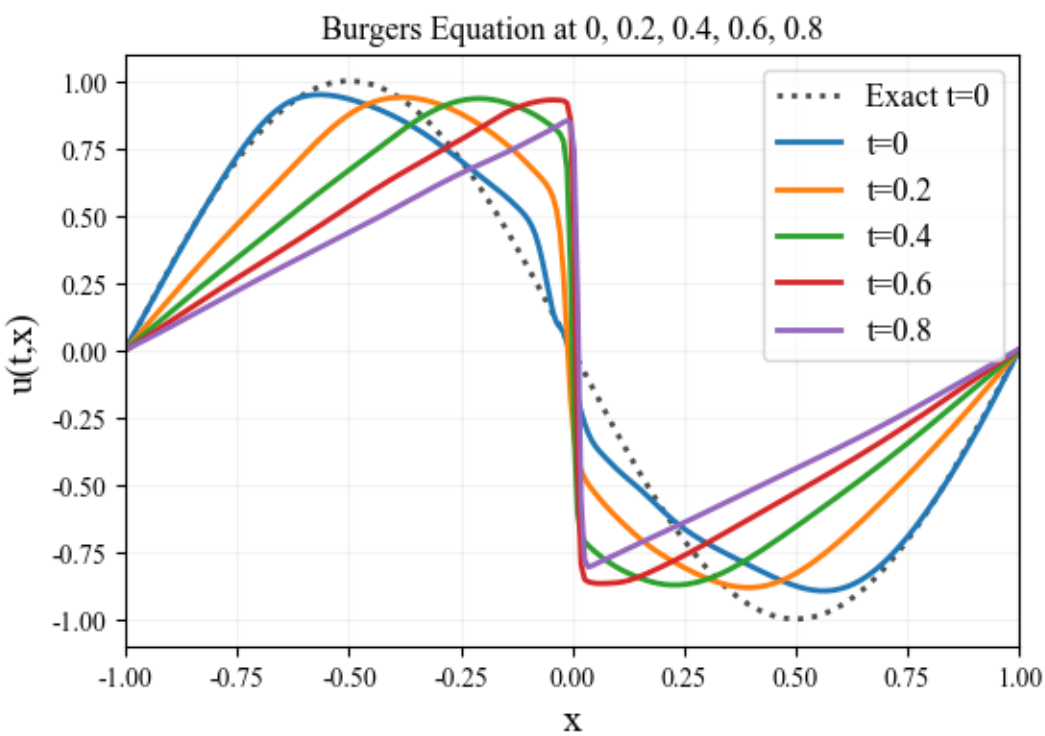
```
epoch 14500, Loss: 1.650e-02
# of training data: 5100
epoch 15000, Loss: 1.446e-02
# of training data: 5200
epoch 15500, Loss: 1.274e-02
# of training data: 5300
epoch 16000, Loss: 1.230e-02
# of training data: 5400
epoch 16500, Loss: 1.063e-02
# of training data: 5500
epoch 17000, Loss: 9.838e-03
# of training data: 5600
epoch 17500, Loss: 1.545e-02
# of training data: 5700
epoch 18000, Loss: 1.379e-02
# of training data: 5800
epoch 18500, Loss: 9.087e-03
# of training data: 5900
epoch 19000, Loss: 1.552e-02
# of training data: 6000
epoch 19500, Loss: 9.306e-03
# of training data: 6100
epoch 20000, Loss: 1.348e-02
# of training data: 6200
epoch 20500, Loss: 8.611e-03
# of training data: 6300
epoch 21000, Loss: 8.567e-03
# of training data: 6400
epoch 21500, Loss: 8.459e-03
# of training data: 6500
epoch 22000, Loss: 8.392e-03
# of training data: 6600
epoch 22500, Loss: 8.125e-03
# of training data: 6700
epoch 23000, Loss: 7.979e-03
# of training data: 6800
epoch 23500, Loss: 7.886e-03
# of training data: 6900
epoch 24000, Loss: 7.812e-03
# of training data: 7000
epoch 24500, Loss: 7.748e-03
# of training data: 7100
epoch 25000, Loss: 7.708e-03
# of training data: 7200
epoch 25500, Loss: 7.620e-03
# of training data: 7300
epoch 26000, Loss: 7.562e-03
# of training data: 7400
```

```
epoch 26500, Loss: 7.509e-03
# of training data: 7500
epoch 27000, Loss: 7.485e-03
# of training data: 7600
epoch 27500, Loss: 7.442e-03
# of training data: 7700
epoch 28000, Loss: 7.535e-03
# of training data: 7800
epoch 28500, Loss: 7.379e-03
# of training data: 7900
epoch 29000, Loss: 1.003e-02
# of training data: 8000
epoch 29500, Loss: 7.294e-03
# of training data: 8100
epoch 30000, Loss: 7.436e-03
Duration time is 352.175 seconds
```

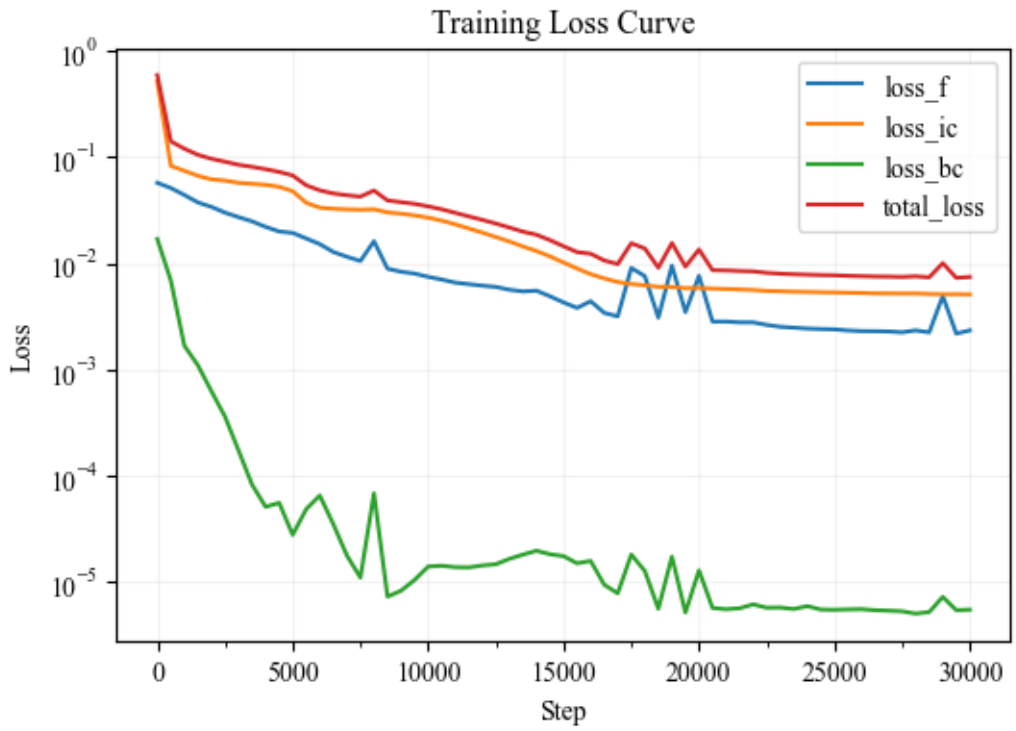
[19]:
```
model_rar = pro_rar[0]

##
plot_sol_2d(model_rar)
plot_sol_3d(model_rar)
plot_sol_curve(model_rar)
```




Solution of Burgers Equation on (x, t) Plane
0.5
0.0
-0.5
-1.0
-0.5
0.0
0.5
1.0
x
0.0
0.2
0.4
0.6
0.8
t

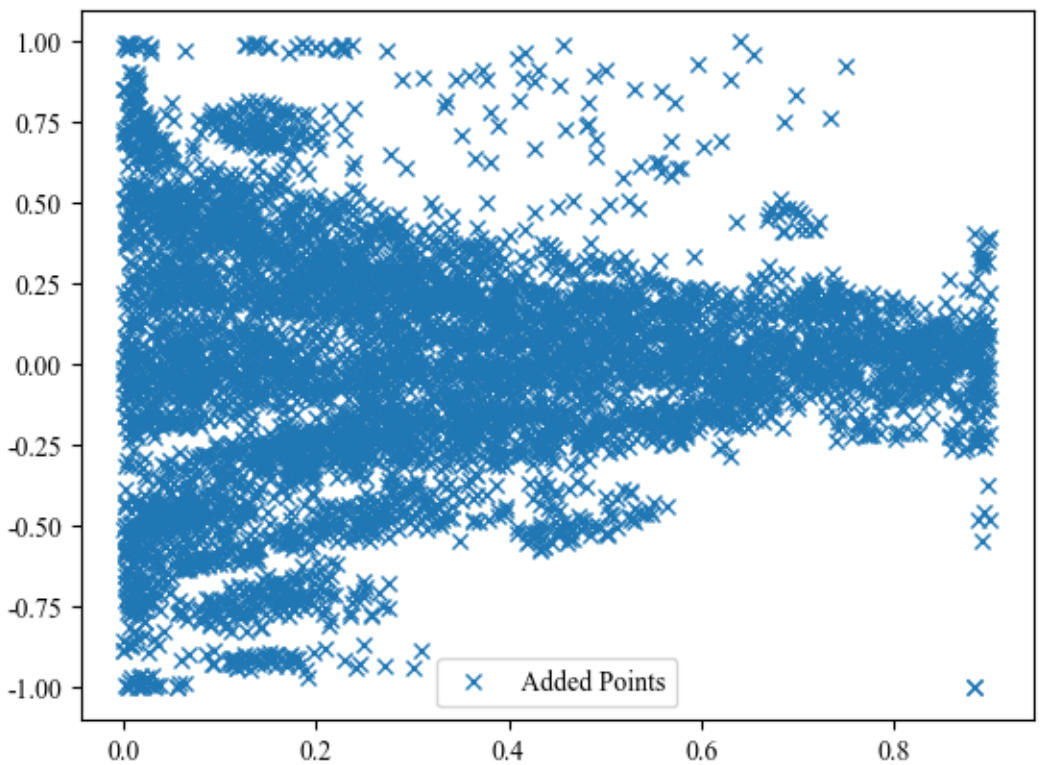

Burgers Equation at 0, 0.2, 0.4, 0.6, 0.8
Exact t=0
t=0
t=0.2
t=0.4
t=0.6
t=0.8
u(t,x)
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
-1.00
-0.75
-0.50
-0.25
0.00
0.25
0.50
0.75
1.00
x

[20]:
```
#

loss_rar = pro_rar[1:-1]
loss_rar_dict = dict(zip(loss_name,loss_rar))

plot_loss_dict(loss_rar_dict)
```



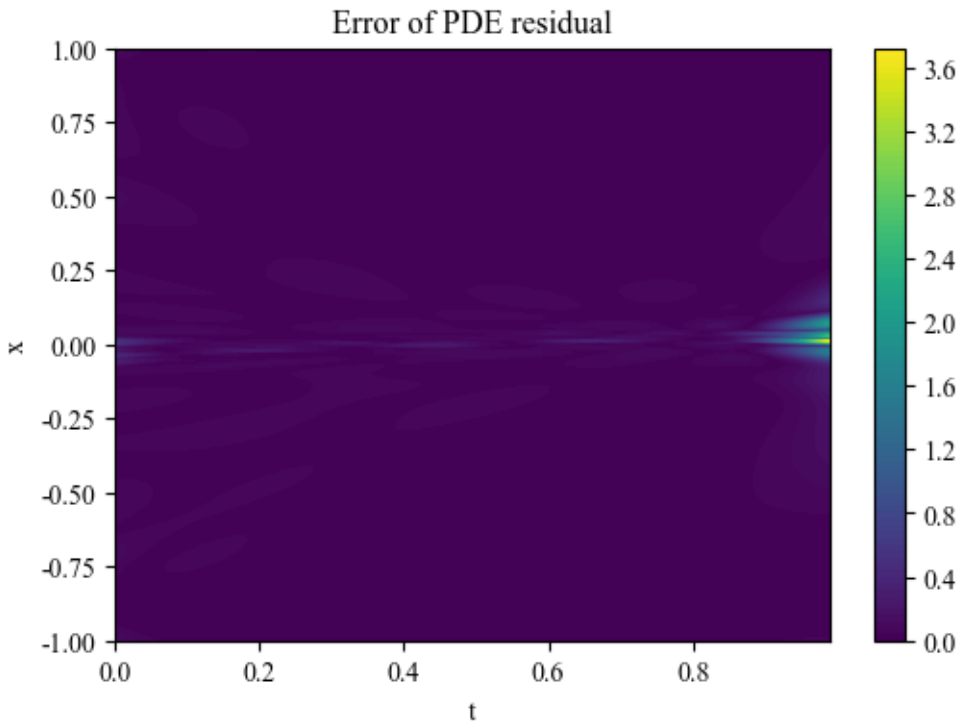
[21]:
```
#

added_x, added_t = pro_rar[-1]['x'], pro_rar[-1]['t']
added_x
print(type(added_x[0]))
plt.figure()
plt.plot(added_t, added_x, marker='x',linestyle='None', label='Added Points')
plt.legend()
plt.show()
```

```
<class 'float'>
```



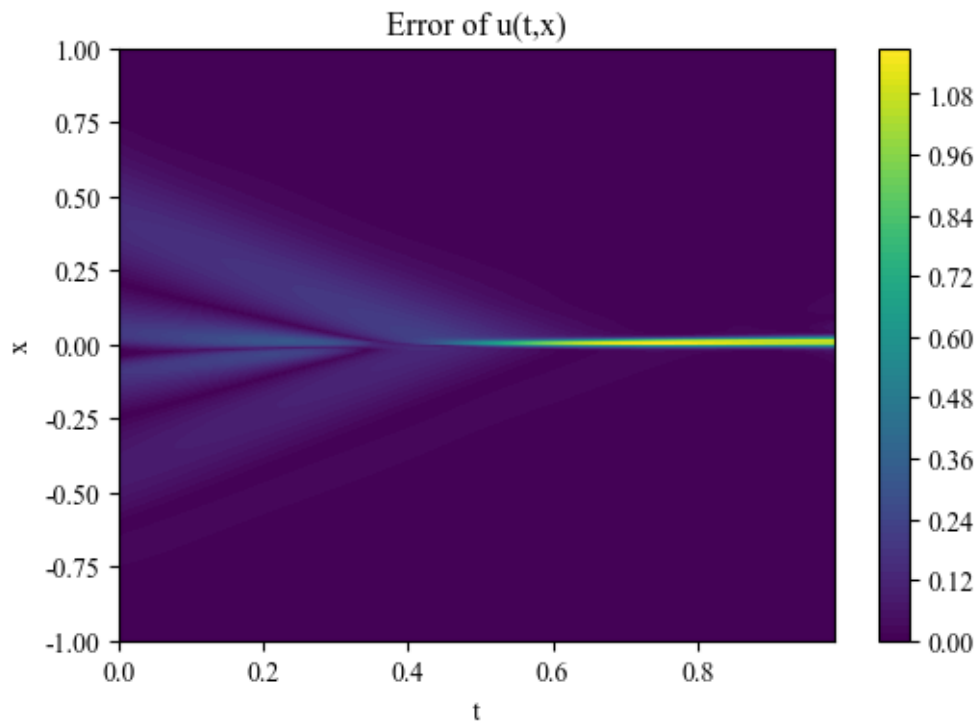
[22]:
```
#

x_test, t_test = base_x, base_t
X_, T_ = np.meshgrid(x_test,t_test)
X_, T_ = X_.reshape(-1,1), T_.reshape(-1,1)

f_pred_rar = model_rar.pdenn(to_tensor(X_),to_tensor(T_)).detach()

mse_pde_rar = mean_squared_error(f_pred_rar, torch.zeros_like(f_pred_rar))
mae_pde_rar = mean_absolute_error(f_pred_rar, torch.zeros_like(f_pred_rar))
print(f'mse_pde:{mse_pde_rar:.5e}, mae_pde:{mae_pde_rar:.5e}')

f_pred_rar = np.abs(f_pred_rar.reshape(len(t_test),len(x_test)).numpy())

plot_sol_2d(x=x_test, t=t_test, u=f_pred_rar,title='Error of PDE residual')
```

```
mse_pde:9.58060e-03, mae_pde:3.07449e-02
```



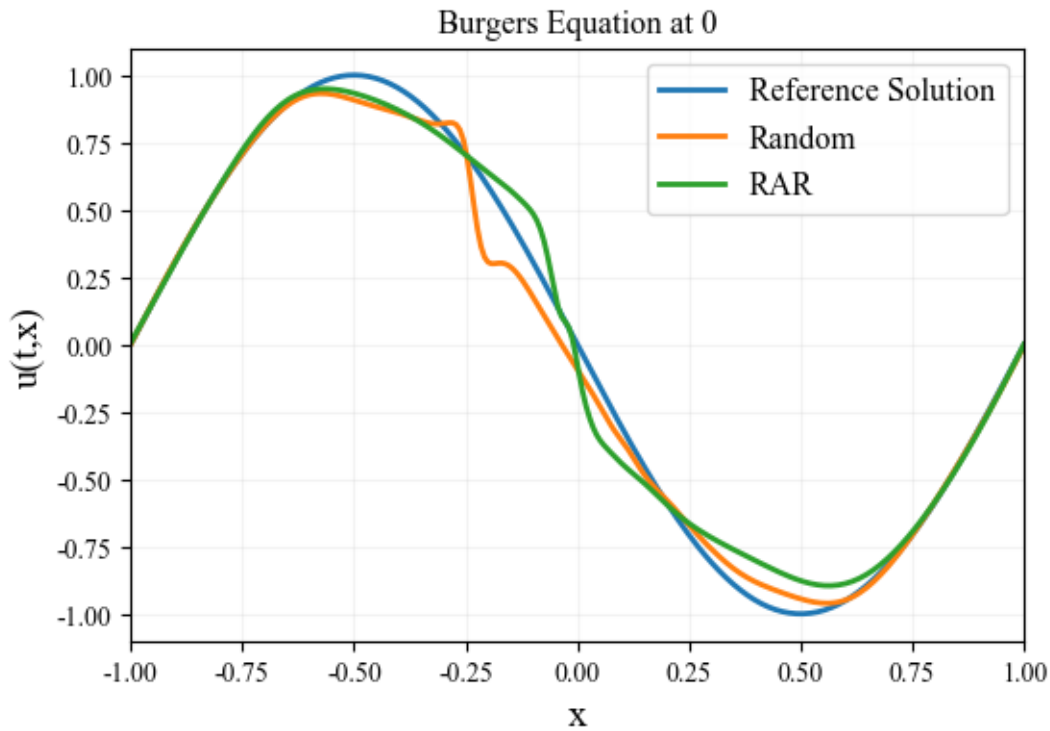
[23]:
```
#
u_rar_pred, mse_rar, mae_rar = evaluate(model_rar)
print(u_rar_pred.shape)
print(f"MSE:{mse_rar:.5e}, MAE:{mae_rar:.5e}")


err_rar = np.abs(u_rar_pred-base_u.T)
plot_sol_2d(x=base_x, t=base_t, u=err_rar, title='Error of u(t,x)')
```

```
(100, 256)
MSE:6.75816e-03, MAE:3.04283e-02
```

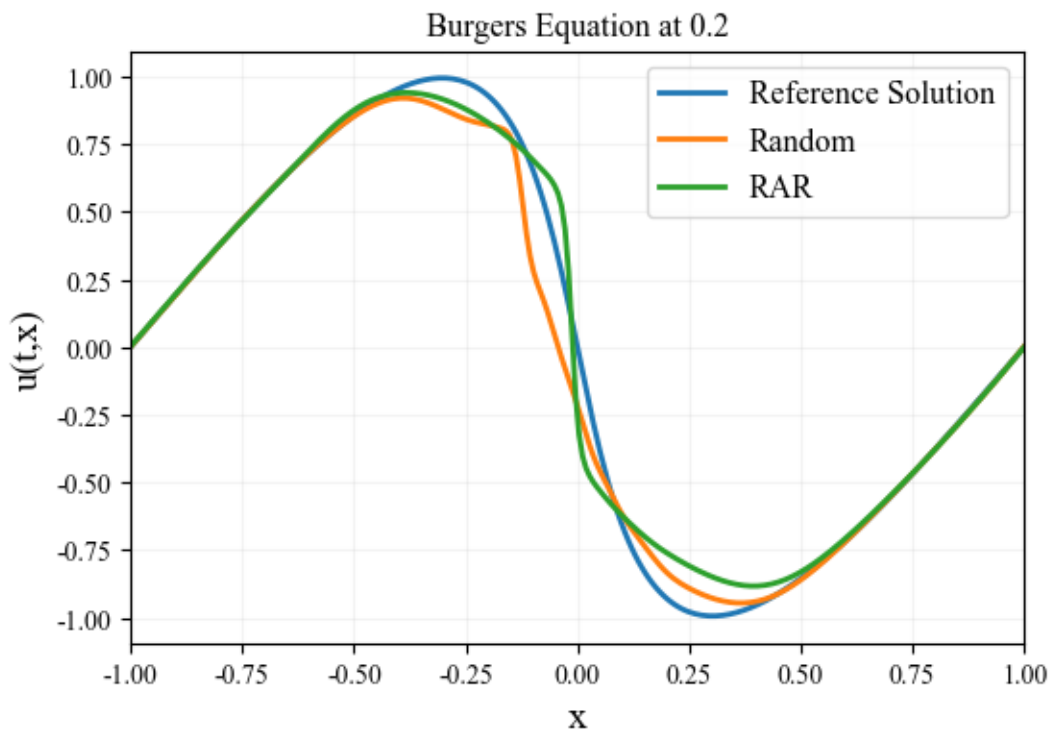


[24]:
```
#

time_slices=[0, 0.2, 0.4, 0.6, 0.8]
for t in time_slices:
    plt.figure(figsize=(6,4))
    plot_sol_curve(x=base_x, t=base_t, u=base_u,␣
 ↪time_slices=[t],label='Reference Solution')
    plot_sol_curve(model=model, time_slices=[t],label='Random')
    plot_sol_curve(model=model_rar,time_slices=[t],label='RAR')
    plt.show()
```

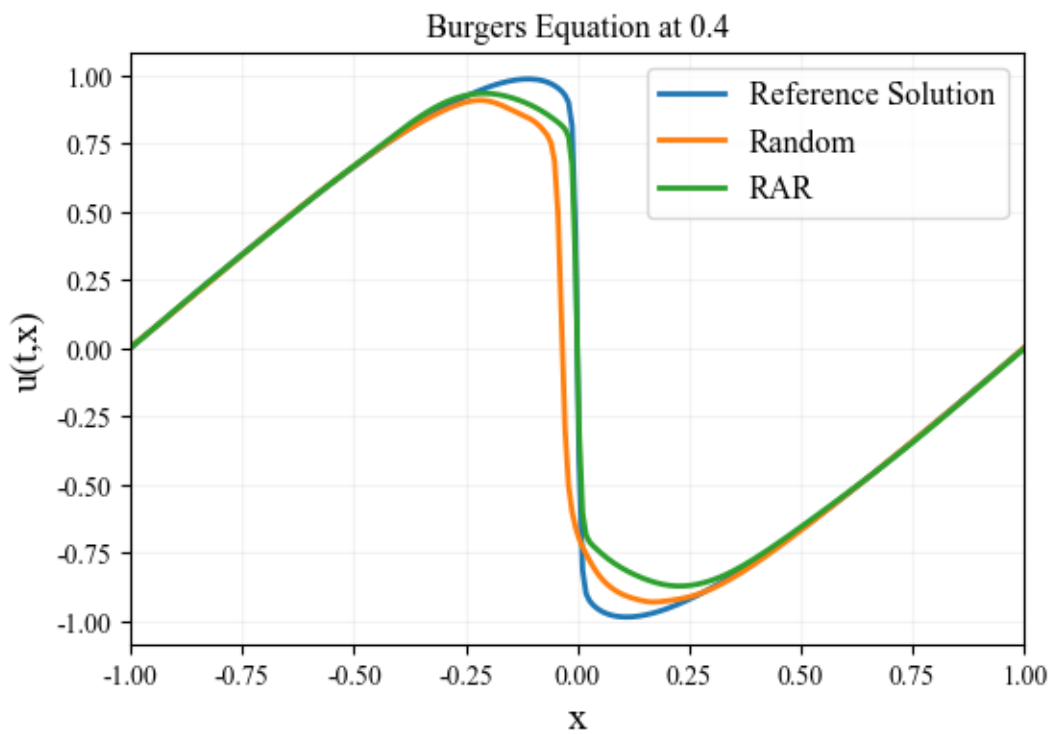
```
(256, 1)
```


Burgers Equation at 0
Reference Solution
Random
RAR
u(t,x)
x
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
-1.00
-0.75
-0.50
-0.25
0.00
0.25
0.50
0.75
1.00

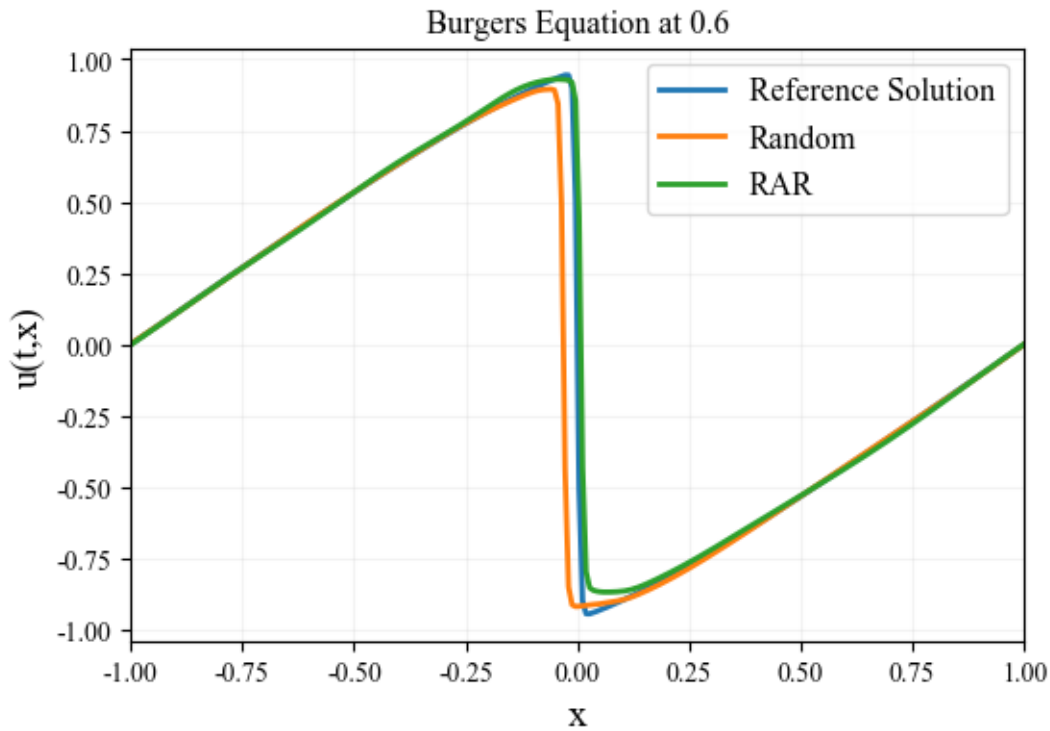
```
(256, 1)
```


Burgers Equation at 0.2
Reference Solution
Random
RAR
u(t,x)
x
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
-1.00
-0.75
-0.50
-0.25
0.00
0.25
0.50
0.75
1.00

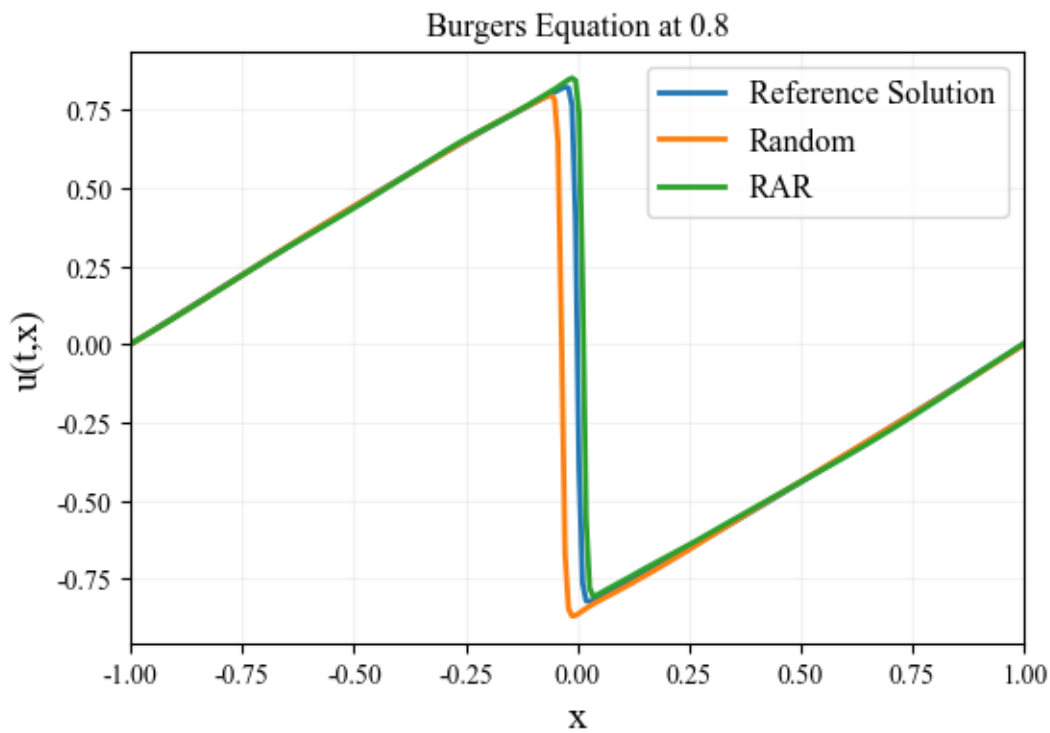
```
(256, 1)
```


Burgers Equation at 0.4
Reference Solution
Random
RAR
u(t,x)
x
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
-1.00
-0.75
-0.50
-0.25
0.00
0.25
0.50
0.75
1.00

```
(256, 1)
```


Burgers Equation at 0.6
Reference Solution
Random
RAR
u(t,x)
x
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
-1.00
-0.75
-0.50
-0.25
0.00
0.25
0.50
0.75
1.00

```
(256, 1)
```


Burgers Equation at 0.8
Reference Solution
Random
RAR
u(t,x)
x
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
-0.75
-0.50
-0.25
0.00
0.25
0.50
0.75
1.00